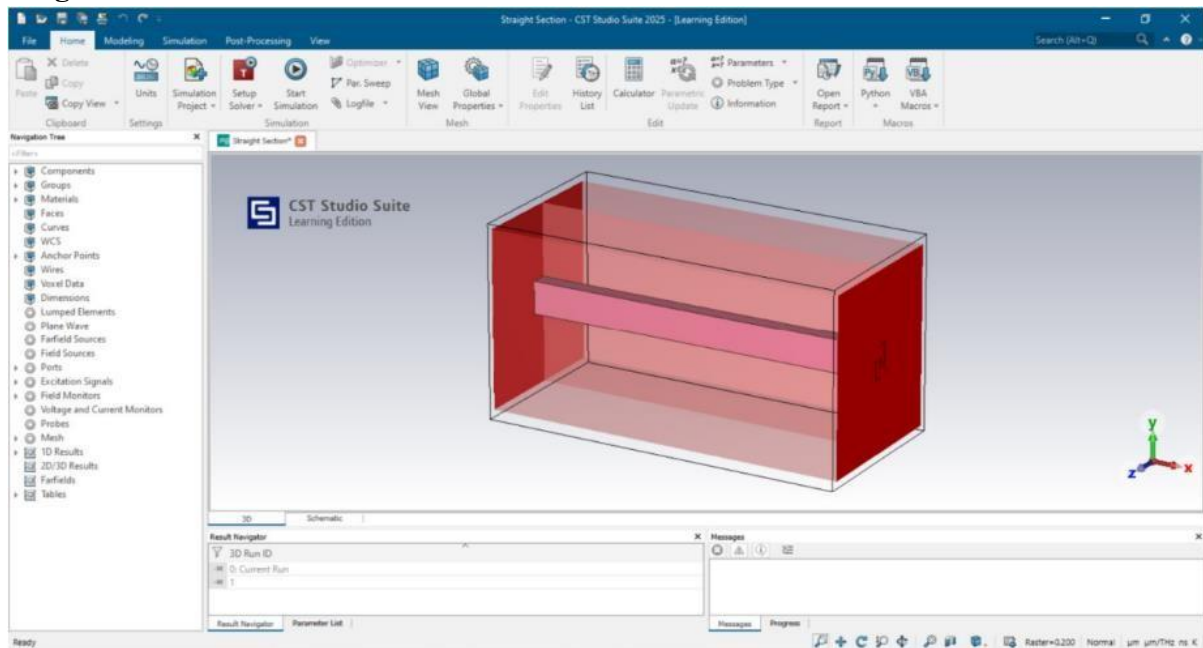


Electromagnetic Field and Power Flow Analysis of a Rectangular Waveguide in CST Studio Suite and determination of S parameters.

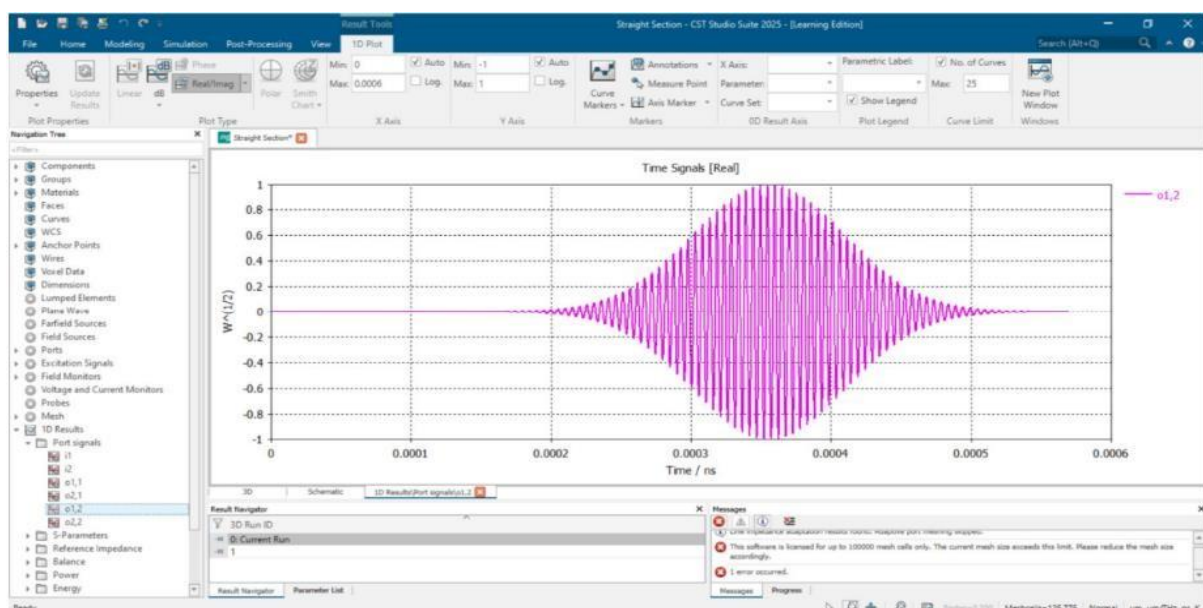
Test Case 1:

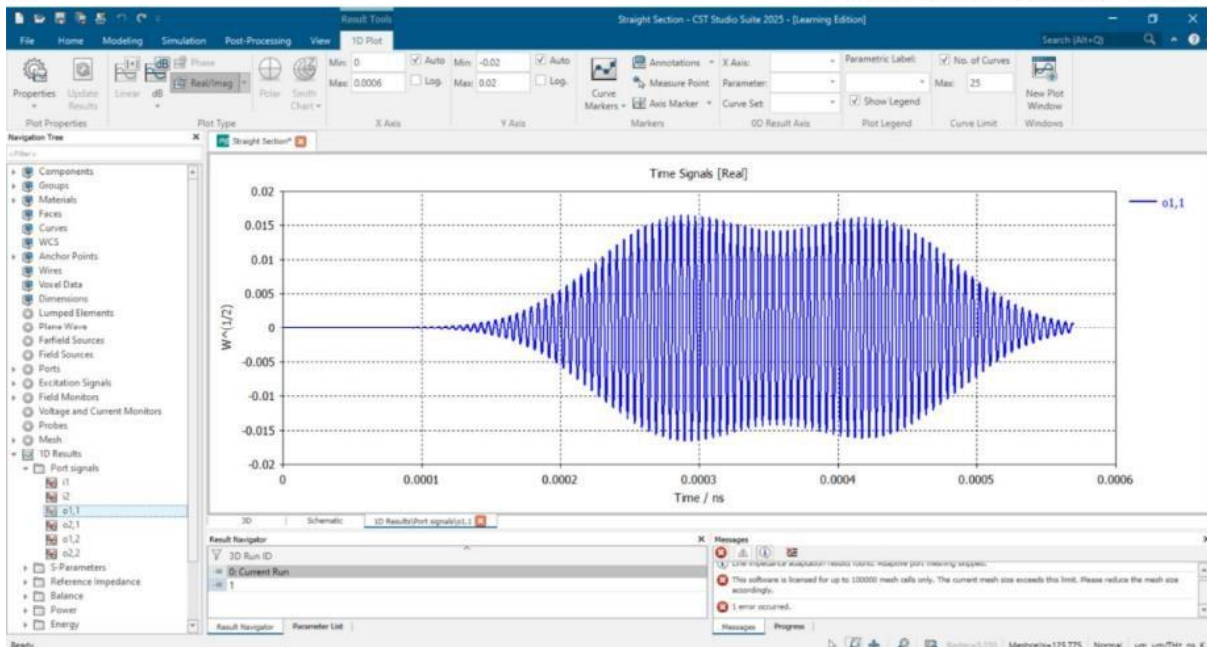
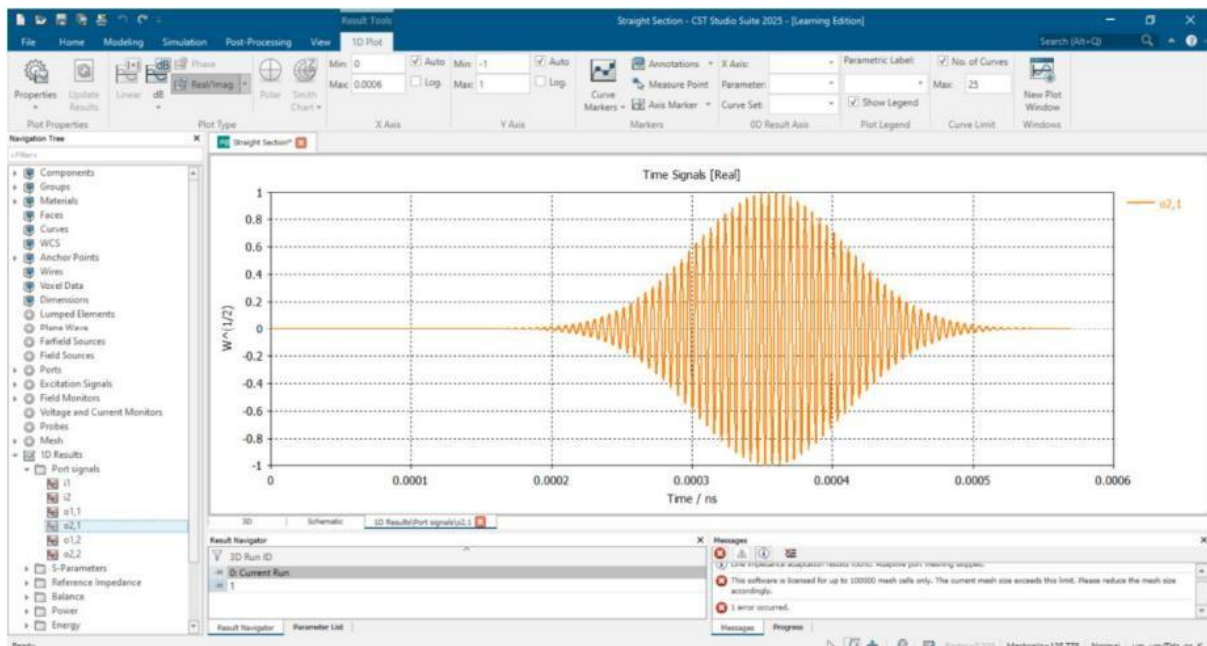
To simulate a Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behaviour, and examine the power distribution among ports over a specified frequency range.



C

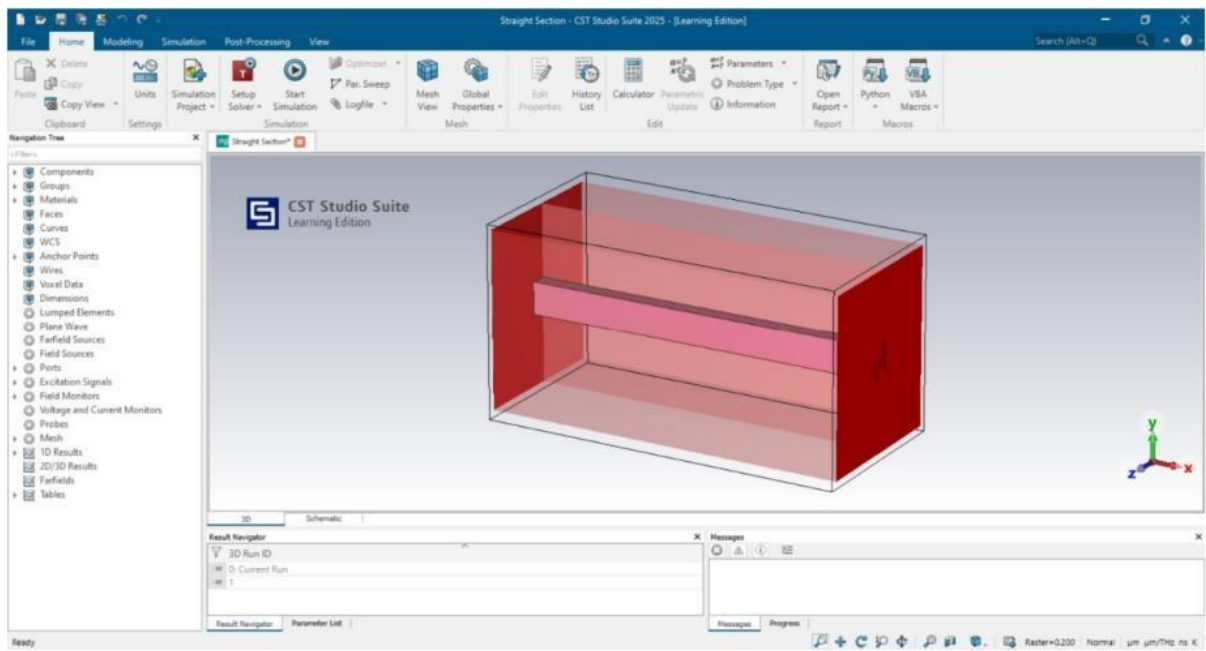
Output:



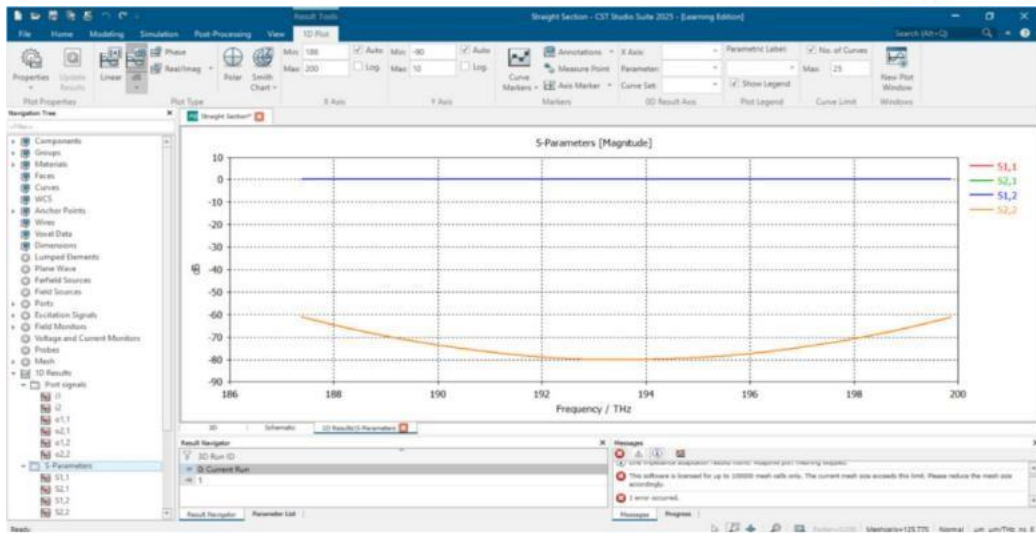
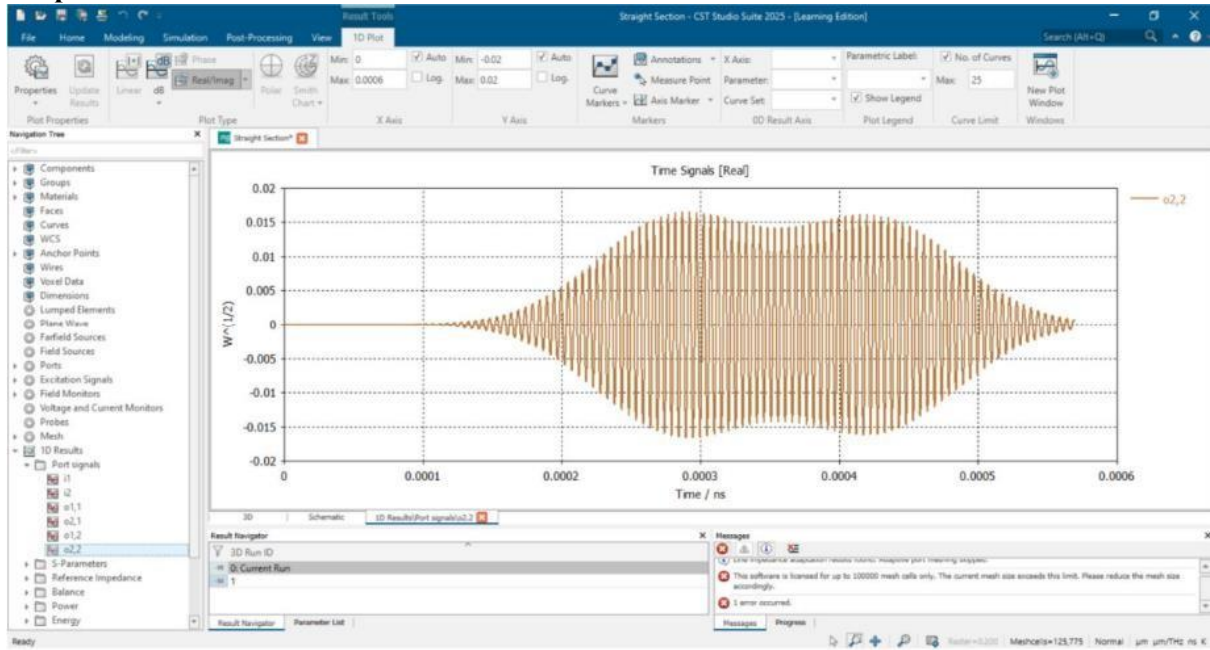


Test Case 2:

To simulate a Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behaviour, and examine the S parameters.



Output:



RESULT:

The rectangular to circular waveguide transition was successfully simulated using CST Studio Suite.

The

electric field distribution at Port 2 was observed and analyzed, showing the expected spatial variation and

field intensity profile across the port region. The magnetic field distribution at Port 2 was also obtained,

indicating the corresponding magnetic field patterns consistent with electromagnetic wave propagation.

The electric and magnetic field parameters at Port 2 were successfully measured and recorded.

The electric and magnetic field parameters at Port 2 were successfully measured and recorded.